



International School

CAPSTONE PROJECT 2

CMU-SE 451

Sign Language Detection System

Submitted by

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PROJECT INFORMATION

Project acronym	SLDS		
Project Title	Sign Language Detection System		
Start Date	15 - Feb - 2024	End Date	20 - May - 2024
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Document Approval

The following signatures are required for approval of this document

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1. Project Overview

1.1. Project Description

Table1: *Project Description*

Project Code	SLDS	Contact Type	Internal Project
Customer	Thuan, Nguyen Tan	End-User	User
Project Type	Internal	Scrum Master	Duy, Huynh Gia
Project Category	Development	Business Domain	Supporting Technology
Application Type	Web		

1.2. Scope and Purpose

1.2.1. Scope

- The project focuses on developing a video call communication system between sign language users and normal people.
- The scope of the project includes building the user interface, integrating real-time translation and gesture recognition technology, as well as ensuring system flexibility and security.

1.2.2. Purpose

- The main aim of the project is to create an effective communication tool between two groups of people with different communication approaches: sign language users and normal people.
- The project aims to reduce communication barriers and enhance the connection and integration of the deaf and mute community into society through technology.

1.3. Assumptions and Constraints

Table 1: *Assumptions and Constraints*

No.	Description	Note
Assumptions		
1	Python version v3.11.0 (or above) and lower version not supported.	Scope
Constraints		

1	The project is completed within 3.5 months	Schedule
2	The project shall conform to the security requirements specified	Security
3	The product operates at a high level of performance	Quality
4	The application operates on the Web	Scope
5	The project will be implemented by a team including 4 members	Resources
6	The financial estimation for the project is at a budget limit of \$4128 Budget	Budget

1.4. Project Objectives

1.4.1. Standard Objective

Table 2: Standard Objectives

Metrics	Unit	Committed	Committed
Start Date	Date	22/01/2024	
End Date	Date	19/05/2024	
Duration	Date	120 days	
Maximum Team Size	Person	4 peoples	
Billable Effort	Person-day	$\$2 * 6 * 4 * 100 = 4800$ (Working hours = Number of working hours per day * Number of members * Number of days. Cost = Working hours * The cost per member per hour. = Working hours * 2)	
Number of work hours per day for one engineer	Person-hour	6	

1.4.2. Specific Objectives

- The deployment system minimizes defects and good control of risks by the project team.
- Done on time, completion of the project early by May 29

- The system operates with high performance and safety for the user. User security data is encrypted and stored carefully, avoiding data loss.
- Apply new technology to the project
- The system is easy to use and user-friendly

1.5. Critical Dependencies

No.	Dependency	Expected delivery date	Note
1	N/a	N/a	
2	N/a	N/a	

1.6. Project Risk

Table 3: Project Risk

Identify Risk	Descriptions	Probability	Impact	Mitigations
Scope definition	Scope changes may arise during the project. The redundant scope may be discovered.	3	4	Analysis and evaluation scope.
Requirement definition	Internal inconsistencies may exist within requirements that may be missing from formal requirement specifications.	2	4	Unifying requirements before proceeding with the analysis.
Resources	Key resources may be unavailable when required. Specific skills may not be available when required.	2	2	Training
Communication	The mentor's requirement may be misunderstood. Project reporting needs may change during the project.	2	3	Having analyzed the requirement, the mentor should review it to ensure that the requirements are not misunderstood.
Health	Health may change during the project. An accident or incident	2	3	The project plan has backup time to ensure that if there are any

	may occur, delaying the project.			health risks, they will not interrupt the project.
Time	The project may not be finished on time.	2	4	Overtime to finish.
Estimate	The size estimate may be below.	3	4	

2. Project development approach

2.1. Agile Methodology

Agile software development refers to a group of software development methodologies based on iterative development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams.

Agile software development is more than frameworks such as Scrum, Extreme Programming, or Feature-Driven Development (FDD).

Agile software development is an umbrella term for a set of frameworks and practices based on the values and principles expressed in the Manifesto for Agile Software Development and the 12 Principles behind it. When you approach software development in a particular manner, it's generally good to live by these values and principles and use them to help figure out the right things to do give your particular context.

2.2. Scrum Process

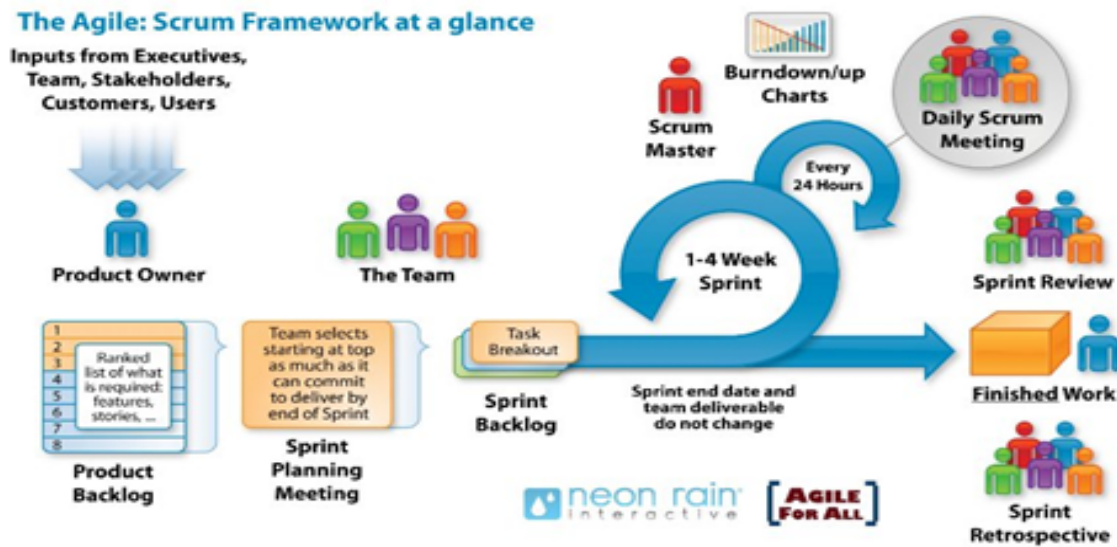


Figure 1: Scrum model

Scrum is a methodology that assists teams in accomplishing project goals promptly and effectively. By implementing Scrum, teams can efficiently utilize their time and resources. When dealing with large projects, Scrum divides them into smaller, more manageable phases known as sprints. These sprints allow for the coding and testing of developments within a specified timeframe.

Scrum is particularly well-suited for fast-paced development projects where adaptability is crucial. The team gains a clear understanding of the project's progress through regular Scrum meetings. As an agile framework, Scrum readily incorporates feedback from both customers and stakeholders, empowering continuous improvement.

The short duration of sprints facilitates easy adjustment and implementation of changes based on the feedback received. Additionally, daily Scrum meetings provide visibility into the individual contributions of each team member.

2.3. Quality Management

2.3.1. Estimates of Defects to be detected

Table 4: *Pre-release review defects*

Process	Planned found by review
Requirement	30
Design	30
Coding	170
Other	45
Total	275

Table 5: *Pre-release test defects*

Process	Planned found by the test
Requirement	28
Design	25
Coding	165
Other	40
Total	258

2.3.2. Quality Control

Table 6: *Quality Control*

Review Item	Type of Review	Reviewer	When
Proposal	Group review		Initial
Project plan, Project schedule, CM Plan	Group review One-person review		End of the Initiation stage
Business analysis and requirements specification document, Use Case catalog	Group review		End of 90% requirements

Design document, object model	Group review		End of 90% design
Stage plans	One-person review		Beginning of each stage
Complex/first specs incl. diagrams time test generated cases, program interactive	Group review		End of detailed design
Code	Group review		After coding for the first few programs

2.3.3. Measurement Program

Table 7: Measurement Program

Data to be collected	Purpose	Responsible	When
Size: No. of KLOC// FP	Calculate the money for the project	PM/SM	At the end of stages
Effort: No. person-day	Predicting the most realistic amount of effort required to develop or maintain software based on incomplete, uncertain, and noisy input	Team members	Daily
Quality: No. defects detected	Identify the root causes of defects and improve the process to avoid introducing defects, which helps to improve the quality of the software product	Reviewer, Tester	Right after the review/test
Schedule	Achieve the goals and priorities in the time available	PM/SM	Weekly and at the end of stages

2.4. Unit Testing Strategy

The system is tested using Unit Testing, a testing technique that uses individual modules tested to determine if there are any problems caused by the developer himself. It is concerned with the functional correctness of the independent modules. The project will be tested by the white box technique of unit testing:

- White Box Testing - used to test each one of those functions' behaviors is tested

Completion criteria:

- All functions in the program are tested.
- All modules in the system are tested.
- All code in the source code must be tested.

2.5. Integration Testing Strategy

- C2SE.05 will be tested by manual testing

2.6. System Testing Strategy

System Testing (ST) is a black-box testing technique performed to evaluate the complete system's compliance with specified requirements. System testing is usually carried out by a team that is independent of the development team to measure the quality of the system unbiased. It includes both functional and Non-Functional testing.

Completion criteria:

- 95% of the test cases must be executed
- Can release when only 5% of test cases remain without affecting the project

3. Estimation**3.1. Size**

Software Scale Drivers	
Precedence	Nominal
Development Flexibility	Nominal
Architecture/ Risk Resolution	Nominal
Team Cohesion	Very High
Process Maturity	Nominal

3.2. Effort

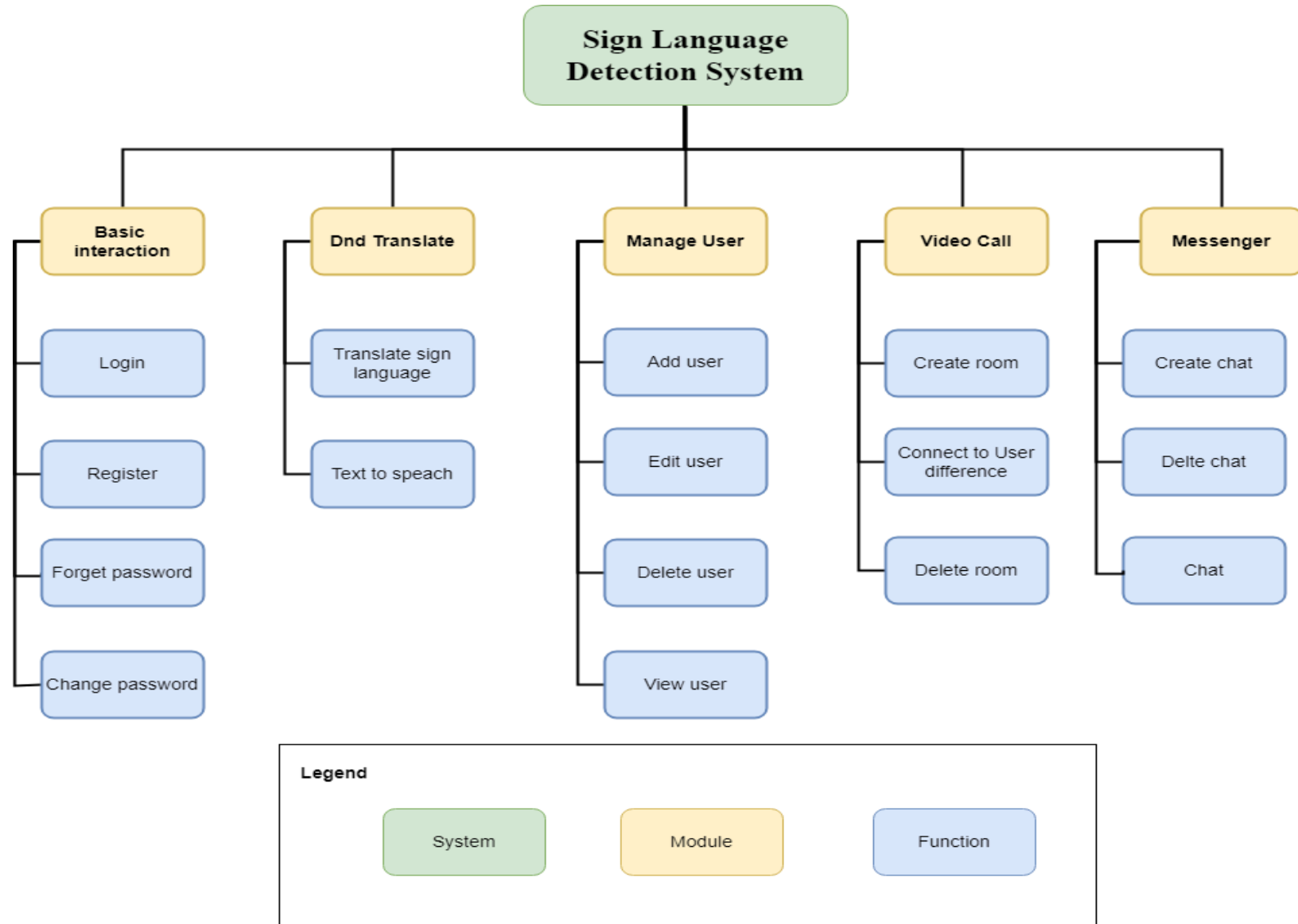
The Effort estimation is documented in C2SE.05-Proposal_V1.0.docx

Table 8: Effort

Activity/ Process	Total budgeted Effort Usage (pd)	Total % budgeted Effort Usage (%)	Sprint 1		Sprint 2		Sprint 3		Sprint 4	
			No.	%	No.	%	No.	%	No.	%
Requirement	\$229.908	7	98.532	3	98.532	3	32.844	1	0	0
Design	\$394.128	12	131.376	4	65.688	2	98.532	3	98.532	3
Coding	\$821.1	25	164.22	5	164.22	5	229.908	7	262.752	8
Unit Testing	\$164.22	5	65.688	2	32.844	1	32.844	1	32.844	1
Testing	\$328.44	10	98.532	3	65.688	2	65.688	2	98.532	3
Deployment	\$426.972	13	131.376	4	65.688	2	98.532	3	131.376	4
Support for Acceptance Test	\$164.22	5	65.688	2	32.844	1	32.844	1	32.844	1
Project Planning	\$164.22	5	65.688	2	32.844	1	32.844	1	32.844	1
Project monitoring	\$164.22	5	65.688	2	32.844	1	32.844	1	32.844	1
Quality Assurance	\$262.752	8	98.532	3	32.844	1	65.688	2	65.688	2
Training	\$164.22	5	65.688	2	32.844	1	32.844	1	32.844	1
Total	\$3284.4	100%		32%		20%		23%		25%

3.3. Schedule

3.3.1. Project Milestone & Deliverables

**Figure 2:** *Project Milestone & Deliverable*

3.3.2. Work Breakdown Structure

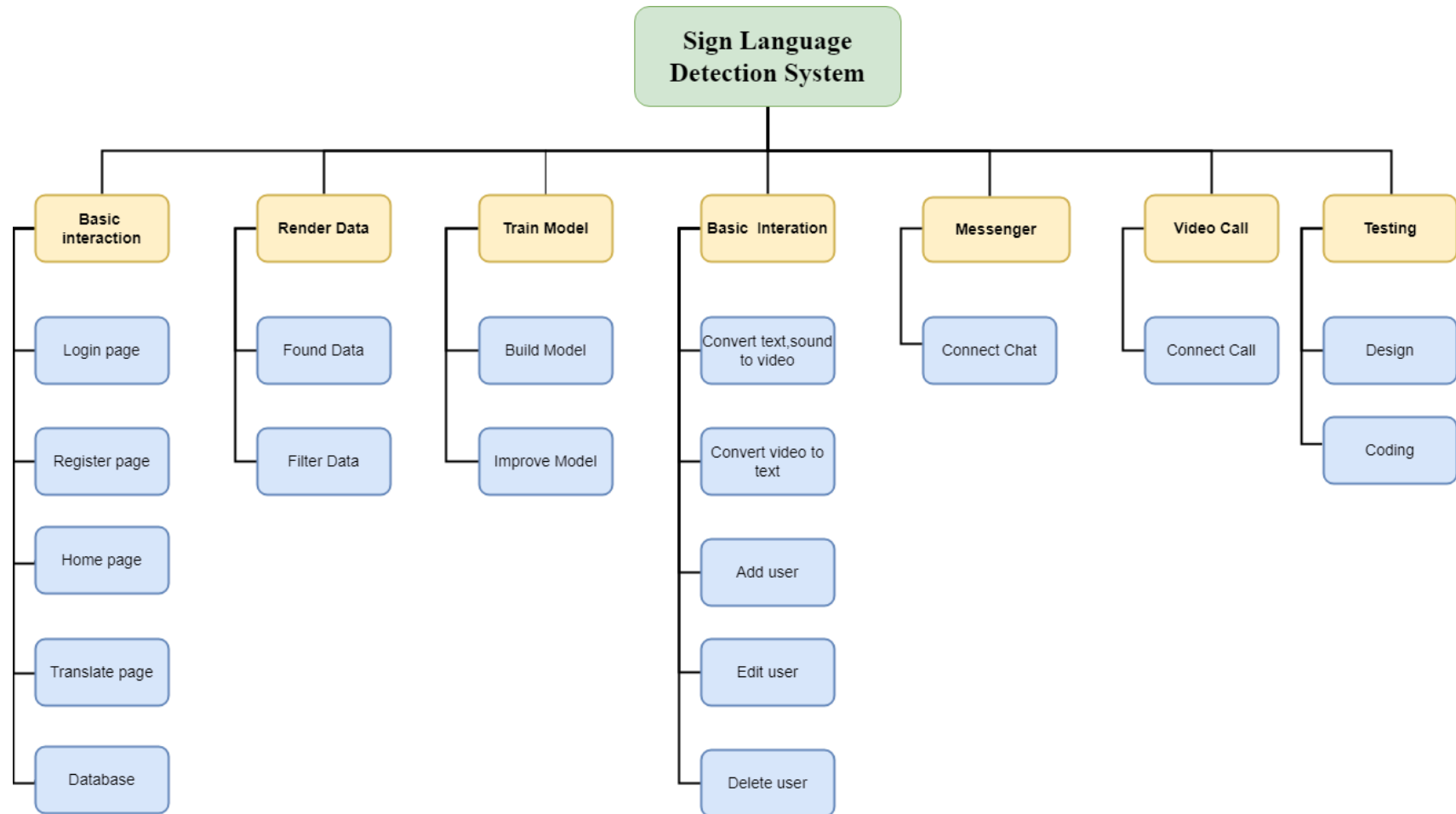


Figure 3: Work Breakdown Structure

3.3.3. Detailed Schedule

No.	Task name	Duration (day)	Start	Finish	Resource names
1	Initial	30	22/01/2024	21/02/2024	Team
1.1	Discuss project ideal	3	22/01/2024	24/01/2024	Team
1.2	Gathering Requirement	3	25/01/2024	27/01/2024	Team
1.3	Create Proposal document	3	28/01/2024	01/02/2024	Team
1.4	Create User Story document	2	02/02/2024	03/02/2024	Team
1.5	Create Product Backlog document	3	04/02/2024	06/02/2024	Team
1.6	Create Architecture document	3	07/02/2024	09/02/2024	Team
1.7	Create Database document	3	10/02/2024	12/02/2024	Team
1.8	Create User Interface document	2	13/02/2024	14/02/2024	Team
1.9	Create Test Plan document	2	15/02/2024	17/02/2024	Team
1.10	Pre-study	2	18/02/2024	19/02/2024	Team
1.11	Project kick off meeting	2	20/02/2024	21/02/2024	Team
2	Development	80	22/02/2024	16/05/2024	Team
2.1	Sprint 1	20	22/02/2024	13/03/2024	Team
2.1.1	Initial Sprint 1	2	20/02/2024	21/02/2024	Team
2.1.1.1	Sprint planning meeting	1	20/02/2024	20/02/2024	Team
2.1.1.2	Create Sprint Backlog for Sprint 1	1	20/02/2024	20/02/2024	Team
2.1.1.3	Create Test Plan document for Sprint 1	1	20/02/2024	20/02/2024	Team
2.1.1.4	Create Database for Sprint 1	2	20/02/2024	21/02/2024	Team
2.1.2	Design UI	3	22/02/2024	23/02/2024	Team
2.1.2.1	[UI] Sign up	1	22/02/2024	22/02/2024	Team
2.1.2.2	[UI] Sign in	1	23/02/2024	23/02/2024	Team
2.1.2.3	[UI] Log out	1	23/02/2024	23/02/2024	Team
2.1.3	Code	6	24/02/2024	29/03/2024	Team
2.1.3.1	[ML] Train Model	1	25/02/2024	25/02/2024	Team
2.1.3.2	[BE] Convert Text To Sign Language (Translate)	1	26/02/2024	26/02/2024	Team
2.1.3.3	[FE/BE] Sign Up	1	27/02/2024	27/02/2024	Team
2.1.3.4	[FE/BE] Sign In	1	28/02/2024	28/02/2024	Team
2.1.3.5	[FE/BE]Logout	1	29/02/2024	29/02/2024	Team

2.1.4	Design Test case	1	06/03/2024	06/03/2024	Team
2.1.5	Testing	2	07/03/2024	08/03/2024	Minh
2.1.6	Fix bug	2	09/03/2024	10/03/2024	Team
2.1.7	Re-testing	2	11/03/2024	12/03/2024	Team
2.1.8	Release Sprint 1	1	13/03/2024	13/03/2024	Team
2.1.8.1	Sprint review meeting	1	13/03/2024	13/03/2024	Team
2.1.8.2	Retrospective	1	13/03/2024	13/03/2024	Team
2.2	Sprint 2	20	14/03/2024	03/04/2024	Team
2.2.1	Initial Sprint 2	2	14/03/2024	15/03/2024	Team
2.2.1.1	Sprint planning meeting	1	14/03/2024	14/03/2024	Team
2.2.1.2	Create Sprint Backlog for Sprint 2	1	14/03/2024	14/03/2024	Team
2.2.1.3	Create Test Plan document for Sprint 2	1	14/03/2024	14/03/2024	Team
2.2.1.4	Update Database for Sprint 2	2	15/03/2024	15/03/2024	Team
2.2.2	Code	11	16/03/2024	27/03/2024	Team
2.2.2.1	[ML] Train Model	1	16/03/2024	16/03/2024	Team
2.2.2.2	[FE/BE] Sign up	1	17/03/2024	17/03/2024	Team
2.2.2.3	[FE/BE] Sign in	1	18/03/2024	18/03/2024	Team
2.2.2.4	[FE/BE] Log out	1	19/03/2024	19/03/2024	Team
2.2.2.5	[BE] Convert Voice To Sign Language (Translate)	2	20/03/2024	21/03/2024	Team
2.2.2.6	[BE] Convert Sign Language To Text And Voice (Translate)	1	22/03/2024	22/03/2024	Team
2.2.2.7	[FE/BE] Create Room Video Call	3	23/03/2024	25/03/2024	Team
2.2.2.8	[BE] Convert Voice To Sign Language (Video Call)	4	24/03/2024	27/03/2024	Team
2.2.4	Design Test case	1	28/03/2024	28/03/2024	Team
2.2.5	Testing	1	29/03/2024	29/03/2024	Team
2.2.6	Fix bug	1	01/04/2024	01/04/2024	Team
2.2.7	Re-testing	1	02/04/2024	02/04/2024	Team
2.2.8	Release Sprint 2	1	03/04/2024	03/04/2024	Team
2.2.8.1	Sprint review meeting	1	03/04/2024	29/04/2024	Team
2.2.8.2	Retrospective	1	03/04/2024	29/04/2024	Team

2.3	Sprint 3	20	04/04/2024	24/04/2024	Team
2.3.1	Initial Sprint 3	2	04/04/2024	05/04/2024	Team
2.3.1.1	Sprint planning meeting	1	04/04/2024	04/04/2024	Team
2.3.1.2	Create Sprint Backlog for Sprint 3	1	04/04/2024	04/04/2024	Team
2.3.1.3	Create Test Plan document for Sprint 3	1	05/04/2024	05/04/2024	Team
2.3.2	Code	9	06/04/2024	17/04/2024	Team
2.3.2.1	[ML] Train Model	1	06/04/2024	06/04/2024	Team
2.3.2.1	[BE] Convert Sign Language To Voice (Video Call)	1	07/04/2024	07/04/2024	Team
2.3.2.2	[FE/BE] Send Email Invite User	1	08/04/2024	08/04/2024	Team
2.3.2.3	[BE] Leave Room	1	09/04/2024	09/04/2024	Team
2.3.2.4	[FE/BE] Search User	1	10/04/2024	10/04/2024	Team
2.3.2.5	[FE/BE] Add User To Conversation	2	11/04/2024	12/04/2024	Team
2.3.2.6	[FE/BE] Add Message	2	13/04/2024	14/04/2024	Team
2.3.3	Design Test case	1	18/04/2024	18/04/2024	Team
2.3.4	Testing	2	19/04/2024	20/04/2024	Team
2.3.5	Fix bug	2	21/04/2024	22/04/2024	Team
2.3.6	Re-testing	2	23/04/2024	24/04/2024	Team
2.3.7	Release Sprint 3	2	18/04/2024	01/04/2024	Team
2.3.7.1	Sprint review meeting	1	18/04/2024	18/04/2024	Team
2.3.7.2	Retrospective	1	18/04/2024	18/04/2024	Team
2.4	Sprint 4	20	25/04/2024	15/05/2024	Team
2.4.1	Initial Sprint 4	2	25/04/2024	26/04/2024	Team
2.4.1.1	Sprint planning meeting	1	25/04/2024	25/04/2024	Team
2.4.1.2	Create Sprint Backlog for Sprint 4	1	25/04/2024	25/04/2024	Team
2.4.1.3	Create Test Plan document for Sprint 4	1	26/04/2024	26/04/2024	Team
2.4.2	Code	12	27/04/2024	08/05/2024	Team
2.4.2.1	[ML] Train Model	1	27/04/2024	27/04/2024	Team
2.4.2.1	[FE/BE] List User	1	28/04/2024	28/05/2024	Team

2.4.2.2	[FE/BE] List Message	3	29/04/2024	01/04/2024	Team
2.4.2.3	[BE] Realtime Messages	1	21/04/2024	21/04/2024	Team
2.4.2.4	[BE] Delete Message	1	22/04/2024	22/04/2024	Team
2.4.2.5	[FE/BE] Create User	1	23/04/2024	23/04/2024	Team
2.4.2.6	[FE/BE] Edit User	2	24/04/2024	25/04/2024	Team
2.4.2.7	[BE] Remove User	1	26/04/2024	26/04/2024	Team
2.4.2.8	[FE/BE] User Detail	2	27/04/2024	28/04/2024	Team
2.4.2.9	[FE/BE] List Conversation	3	29/04/2024	01/05/2024	Team
2.4.3	Design Test case	1	09/05/2024	10/05/2024	Team
2.4.4	Testing	2	11/05/2024	12/05/2024	Team
2.4.5	Fix bug	2	13/05/2024	14/05/2024	Team
2.4.6	Re-testing	1	15/05/2024	15/05/2024	Team
2.4.7	Release Sprint 4	1	16/05/2024	17/05/2024	Team
2.4.7.1	Sprint review meeting	1	16/05/2024	16/05/2024	Team
2.4.7.2	Retrospective	1	17/05/2024	17/05/2024	Team
3	Project's Retrospective Meeting	1	18/05/2024	18/05/2024	Team
4	Final Release	1	19/05/2024	19/05/2024	Team

3.4. Project Schedule

The detailed project schedule is available here. The Project Manager weekly updates the Project Schedule.

Table 9: Project Schedule

No.	Activity	Start date	Responsible	Note
Defect Prevention				
	Sprint 1	22/02/2024	All member	
	Sprint 2	14/03/2024	All member	
	Sprint 3	04/03/2024	All member	

	Sprint 4	25/03/2024	All member	
Quality Control				
	Review: Work Product 1			
	Review: Work Product 2			
	Review: Work Product 3			
Project Tracking				
	<Stage name> milestone review meeting			
	<Stage name> milestone review meeting			
QA				
	Final Inspection: Deliverable 1			
	Final Inspection: Deliverable 2			

3.5. Resource

- Specified as in the section [Project Team](#)

3.6. Infrastructure

Table 10: Infrastructure

Work/Product	Purpose	Expected Availability by	Note
Development Environment			
Microsoft Windows	Operating System	Initiation stage	
Postgresql	Database	Initiation stage	
Brython, SDK	Frameworks that support development	Initiation stage	
JavaScript	The programming language used for development	Initiation stage	
Other Tools			
GitHub	Source version control	Definition stage	
Trello	Task tracking	Initiation stage	

Software			
10 GB space on the Computer	Develop software	Initiation stage	
Draw.io/ Figma	Design	Initiation stage	

3.7. Finance

Table 11: Total Cost

No	Criteria	Price	Amount	Total (USD)
1	Working hours	\$6	2404	\$4080
2	Party	\$4	3	\$48
Total cost				\$4128

Table 12: Cost Description

Description	Amount	Unit
Number of members	4	Person
Number of working-day per week	6	Hours
Number of working-hour per week	5	Days
The duration of the project	120	Days
Party cost per member per time	2	USD

3.8. Training Plan

Table 13: Training plan

Training Area	Participants	When Duration	Waiver, Criteria
Technical			
Python	All members	7 Days	Mandatory
NextJs, ReactJs	All members	7 Days	Mandatory
Process			
Configuration management	All members	2 hours	If already trained
Group review	All members	4 hours	Mandatory

Git, Trello	All members	2 hours	Mandatory
Agile	All members	2 hrs.	Mandatory

4. Project Organization

4.1. Organization Structure

Table14: *Organization Structure*

Role	Responsibility	Name
Scrum Master	- Communicate the value of Scrum - Teach the organization on Scrum to maximize business value - Facilitate Sprint Planning, Daily Scrums, Sprint Reviews, and Retrospective Meetings - Create the Task Board and Sprint Burn down Chart at the start of every Sprint - Attend all Scrum meetings - Preserve the integrity and spirit of the Scrum framework - Maintain the focus of the Team - Make the Team aware of impediments and facilitate efforts to resolve them - Serve as a coach and mentor to members of the Team - Respectfully hold the Team, Product Owner, and Stakeholders accountable for their commitments - Continually work with the Team and business to find and implement improvements	Hien, Tran Minh
Developer	- Achieve the goals defined for each sprint - Attend daily scrum meetings, reporting the planned tasks for each day - Assure to Product Owner and Scrum Master that the allocated work is being performed as planned - Ensure a clear understanding of epics and personas - Provide inputs on the creation of User Stories to the Product Owner - Understand the User Stories	All members
Tester	- Understanding, implement and updating the Agile Test Strategy Work with Product Owners to define Acceptance Criteria and the Definition of Done - Measuring and reporting test coverage across all applicable coverage dimensions. - Configuring, using, and managing test environments and	All members

	test data Writing and executing automated checks and reporting back to the team	
Mentor	- Guide on the process. - Monitoring all activities of the Team. - Help with anything. - Reviews project documents - Review product	Duc, Nguyen Manh

4.2. Project Team

Table15: *Project Team*

Full Name	Position
Duc, Nguyen Manh	Mentor
Hien, Tran Minh	Scrum Master, Dev-team
Dinh, Le Nguyen Nhut	Dev-team
Hiep, Pham Minh	Dev-team
Minh, Huynh Tan Quang	Dev-team

5. Communication & Reporting

5.1. Reporting

Table 16: *Reporting*

Audience / Attendees	Topic / Deliverable	Frequency	Method
- Product Owner - Scrum Master - Team Members	Project Progress Review	Weekly	Google Meet, Zalo
- Product Owner - Scrum Master - Team Members	Explicit Requirement	When needed	Google Meet, Zalo
- Mentor - Scrum master - Team members	Milestone review	End of each Milestone	Google Meet, Face to face
- Scrum master - Team members	Daily tasks	Each day	Messenger, Google Meet

5.2. Communication

Table 17: Communication

Communication Type	Method /Tool	When	Information	Participants /Responsible
Project Task Tracking				
Task scheduling	Trello	At the beginning of every stage, and weekly Refinement and rescheduling as necessary	Planning tasks for each sprint	Project Mgr.(s)
Task assignment	Trello	Weekly	Assign tasks to each member	Project leader Technical
Project Meeting				
Kick-off Meeting	Google Meet	Initiation stage	Project introduction; Project plan review; Risk identification; Obtainment of commitment of relevant stakeholders	Project Mgr.(s), Project Senior Manager, Project Team Members, QA
Project Progress Review Meetings	Face to face, Google Meet	Weekly event on event	Communicate project status; Communicate and resolve any open issue, risks, and changes; Discuss any suggested improvement	Project Mgr.(s), Project Team Members
Milestone Meetings	Face to face, Google Meet	Before milestones	Project objective review, evaluate project performance (quality, schedule, effort), Causal analysis, update project plan for next stage	Project Mgr.(s), Project Senior Manager, Project Team Members, QA

6. Security Aspects

- The credential data is carefully secured by multi-layer encryption. Regularly backup system data.
- Research on network attack prevention solutions to ensure data security, and avoid being exploited and stolen data by hackers.
- Deploy project architecture with a high priority in security. Optimized architectural solutions enable the deployment of data security with 99% reliability.
- Social media, sharing, and use of data must be approved by the end user and verified by the organization's management.

7. References

No.	References	Document Information
1	Scrum Model	https://en.wikipedia.org/wiki/Scrum_(software_development)
		https://www.atlassian.com/agile/scrum
		https://www.digite.com/agile/scrum-methodology/
		https://www.scrum.org/resources/scrum-guide
2	Technical	https://docs.arduino.cc/
		https://www.java.com/en/
		https://developer.android.com/studio

3	Software Engineering Standards	https://www.nws.noaa.gov/oh/hrl/developers_docs/General_Software_Standards.pdf
		https://standards.ieee.org/standard/12208-2017.html
		https://sw-eng.larc.nasa.gov/

8. Definitions and Acronyms

Acronym	Definition	Note
PM	Project Manager	
PTL	Project Technical Leader	
QA	Quality Assurance Officer	
CC	Infrastructure Configuration Controller	
DV	Developer	
URD	User Requirement Document	
SRS	Software Requirement Specification	
ADD	Architecture Design Document	

DDD	Detail Design Document	
TP	Test Plan	
TC	Test Case	
SC	Source Code	
CM	Configuration Management	
CSCI	Computer Software Configuration Items	
CI	Configuration Item	
CCB	Change Control Board	